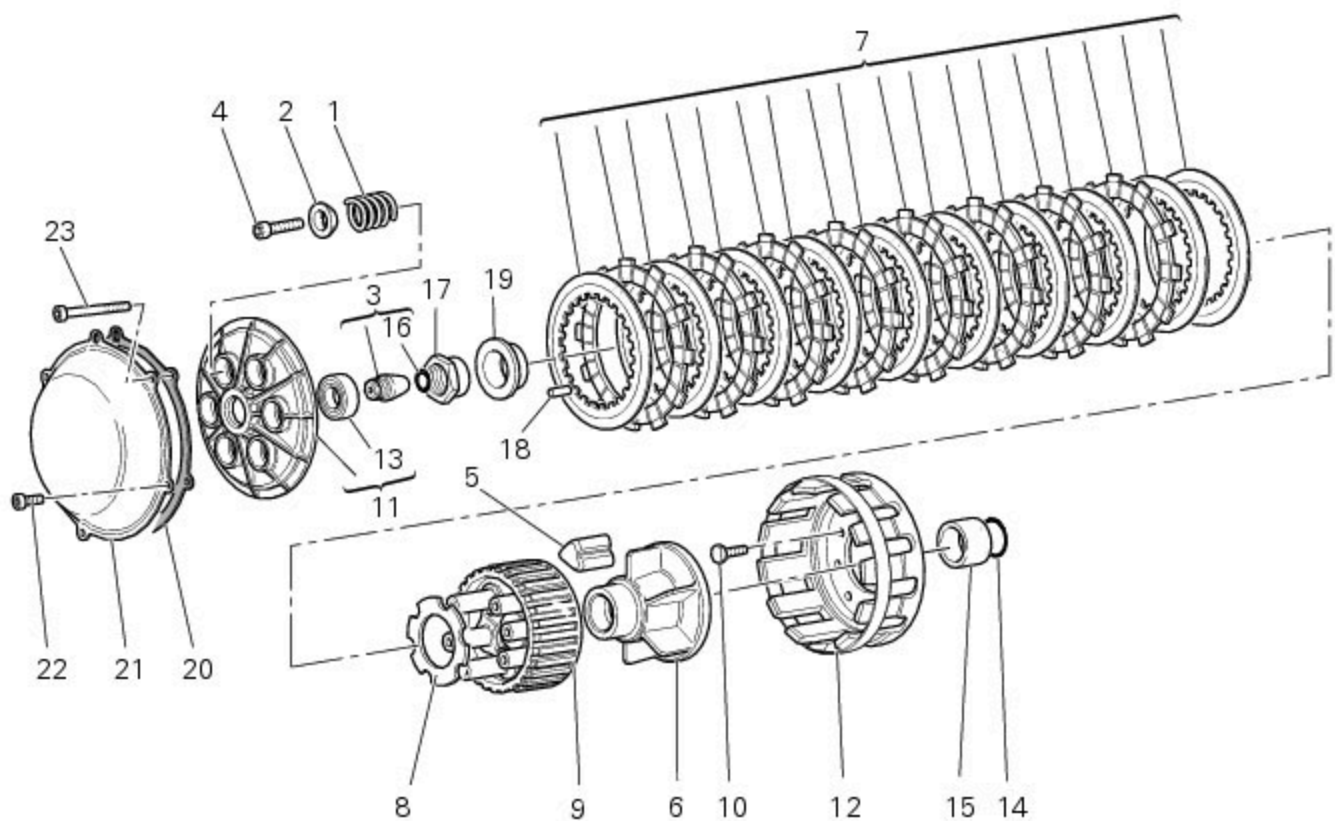


6.1 -Clutch assembly: clutch



- 1 Clutch spring
- 2 Collar
- 3 Clutch lifter
- 4 Screw
- 5 Rubber pad
- 6 Cush drive bushes hub
- 7 Clutch plates
- 8 Thrust washer
- 9 Clutch centre
- 10 Screw
- 11 Pressure plate
- 12 Clutch drum
- 13 Bearing
- 14 O-ring
- 15 Spacer
- 16 O-ring
- 17 Nut
- 18 Pin
- 19 Bush
- 20 Clutch cover gasket
- 21 Clutch cover
- 22 Screw
- 23 Screw

1100 ABS	CLUTCH
1100 ABS	CLUTCH-SIDE CRANKCASE COVER
1100 S ABS	CLUTCH
1100 S ABS	CLUTCH-SIDE CRANKCASE COVER

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

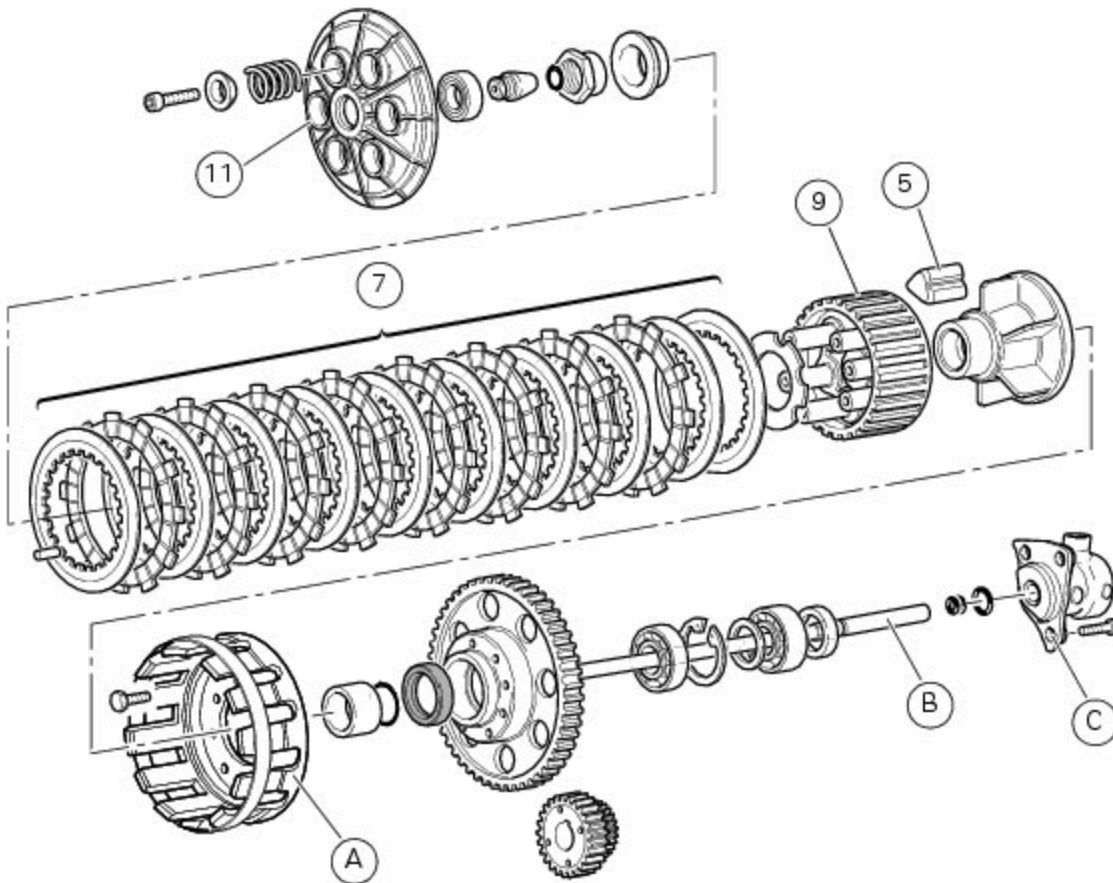
Description of the clutch assembly

The clutch is disengaged by a transmission unit consisting of a thrust piston (C) accommodated inside a small cap mounted to the generator cover. This piston (C) pushes a pushrod (B), which runs through gearbox primary shaft and operates the pressure plate (11) located on top of the clutch plate pack (7).

Motion is transmitted from the clutch centre (9) to the primary shaft through special cush drive rubbers (5), which make for smoother clutch engagement and thus prevent damage to transmission parts.

Accommodated in the clutch housing is a set of drive and driven plates (7). When the clutch is operated, the driven plates push away the drum (9) which is splined onto the gearbox primary shaft.

Before working on the internal clutch parts, check that the clutch operates correctly. Then deal with the problem in a systematic manner.



The following is a list of possible causes of clutch malfunctions.

A clutch which does not disengage may be caused by:

- excessive play of the control lever;
- distorted clutch plates;
- incorrect spring tension;
- faulty clutch release mechanism;
- excessive wear of the hub or clutch drum.

A clutch which slips may be caused by:

- insufficient play of the control lever;
- worn clutch plates;

- weakened springs;
- faulty clutch release mechanism;
- excessive wear of the hub or clutch drum.

A noisy clutch may be caused by:

- excessive backlash between the primary drive gears;
- damaged primary drive gear teeth;
- excessive play between friction plate tabs and the clutch drum;
- worn gear/clutch drum bearings;
- worn cush drive rubbers;
- the presence of metal particles (filings) on the gear teeth.

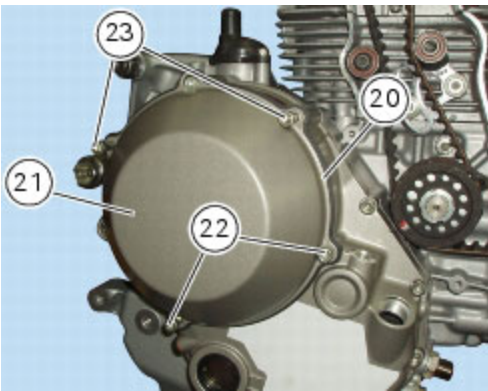
Removal of the clutch



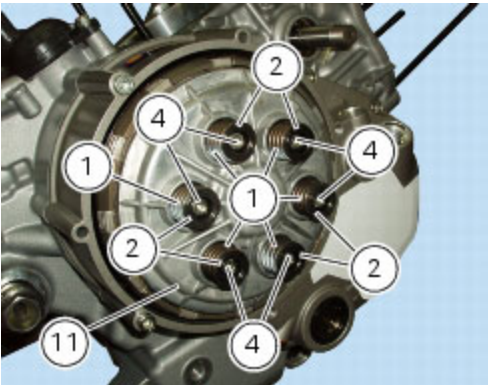
Note

For clarity, the figures show the engine removed from the frame.

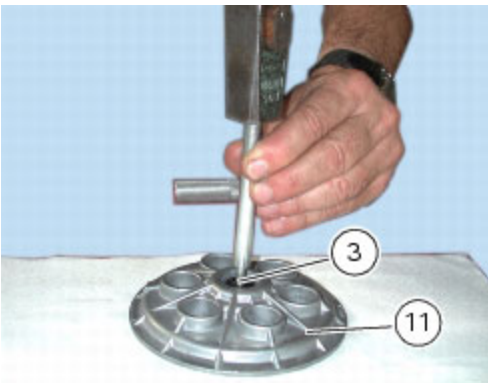
Unscrew the four retaining screws (22) and the two retaining screws (23) from the clutch inspection cover (21). Remove the cover and the sound deadening gasket (20).



Unscrew the retaining screws (4) and remove the collars (2) and the clutch springs (1). Remove the pressure plate (11).



Remove the clutch lifter (3) and its O-ring (**16**) using a suitable drift and a plastic headed mallet.

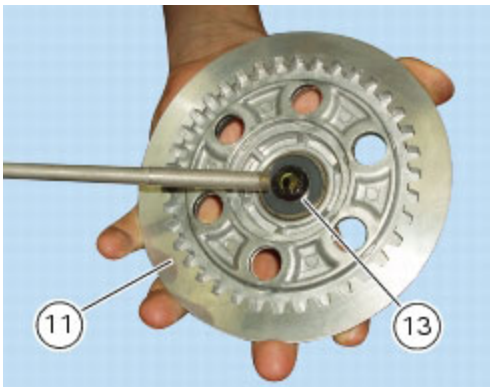
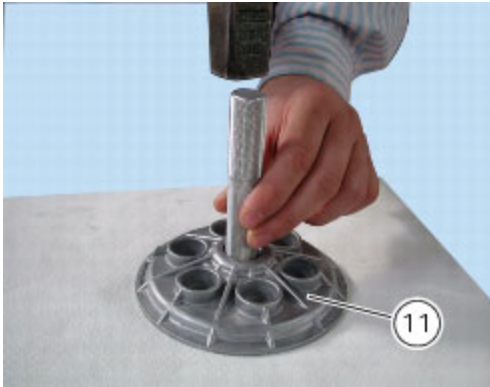


Then remove the bearing (**13**) installed from inner side of the pressure plate (**11**) using a drift of suitable size.



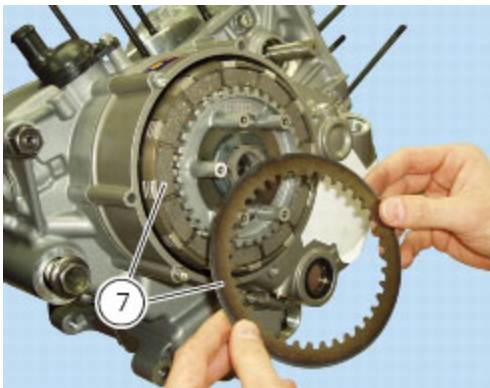
Note

Once disturbed, the bearing must be renewed.

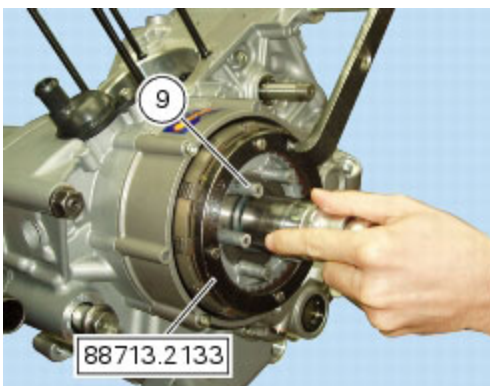


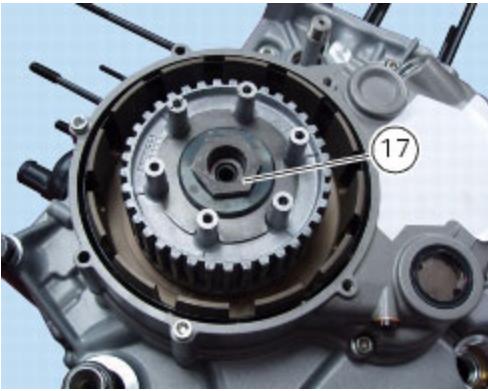
Remove the clutch plates (**7**).

When removing the clutch plates, keep them together in pairs in the correct order and set them aside, tying them together, if necessary.

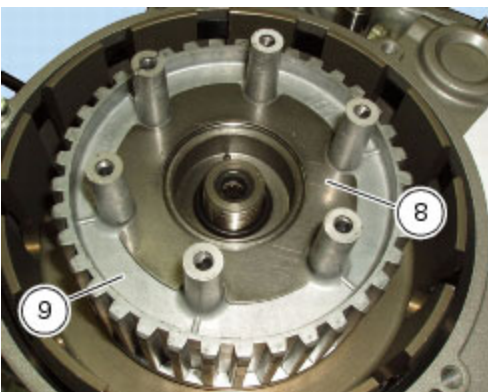
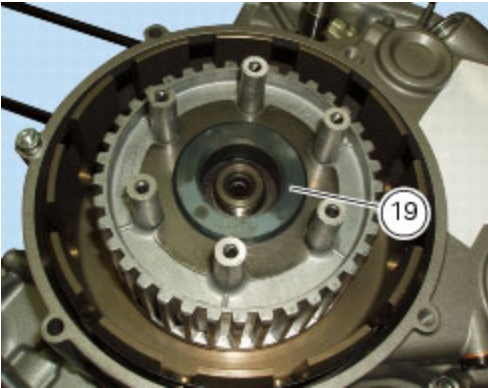


Restrain the clutch centre (**9**) using service tool **88713.2133** and remove the centre nut (**17**).



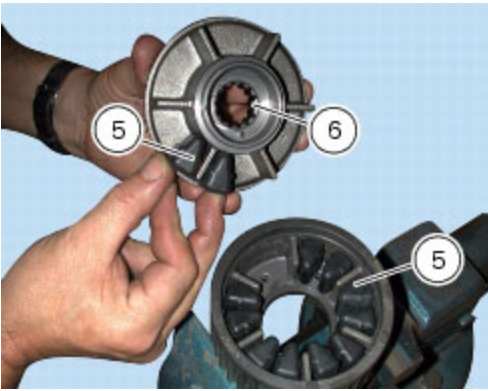


Remove the bush (19) and the washer (8).
Remove the clutch centre (9) complete with the cush drive rubbers from the shaft.



When renewing the cush drive bushes (5), use a drift to remove the hub (6) from the clutch centre (9) and overcome the resistance of the rubbers.





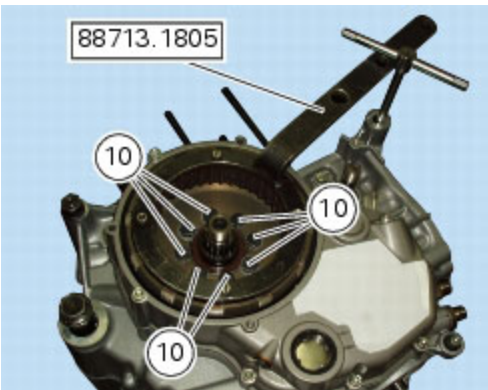
Locate the service tool **88713.1805** to prevent the clutch drum from rotating when loosening the screws (10).



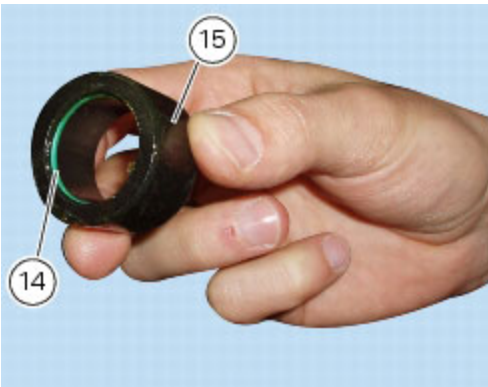
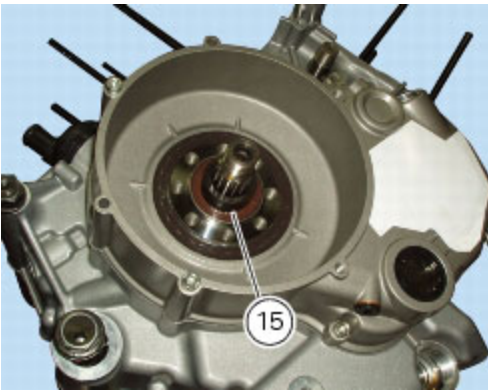
Note

If no cover parts need to be serviced, leave the clutch drum assembly and the primary drive gear installed on the clutch cover.

Undo the eight screws (10) securing the clutch drum to the primary drive gear.
Remove the clutch drum.



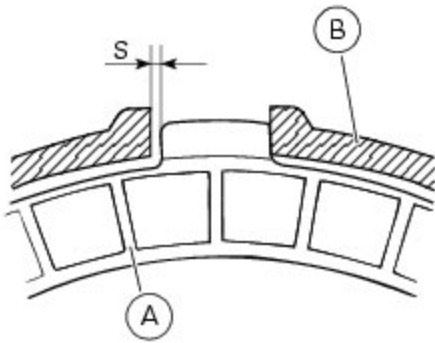
Slide off the spacer (15), taking care not to damage its internal O-ring (14).
Visually check the O-ring for wear and renew it if necessary.



Checking and overhauling the components

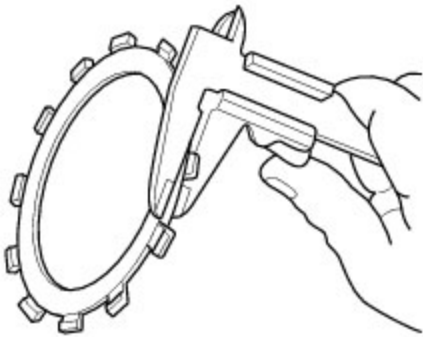
Clearance between the clutch drum and friction plates

Insert the friction plate (A) in the clutch drum (B) and measure tang-to-slot clearance (S) with a feeler gauge. Clearance "S" must not exceed **0.6 mm**. If it does, renew the plates and, if necessary, the clutch drum.



Overhaul of the clutch plates

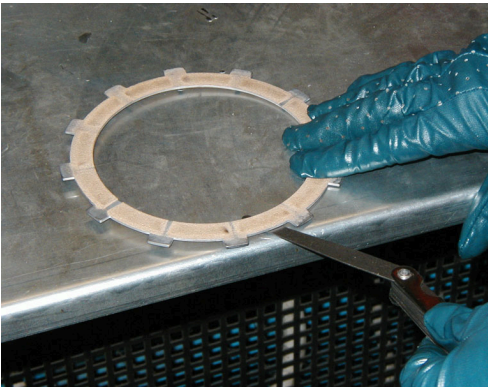
The clutch plates must not show any signs of blackening, grooves or deformation. Measure the thickness of the friction plates. it should not be less than **2.6 mm**.



Important

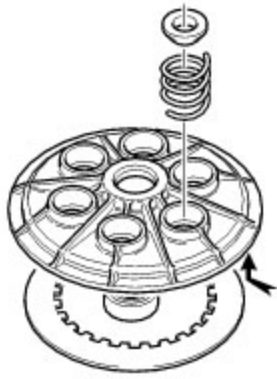
The total thickness of the discs pack must not be less than **46.1 mm**.

Place the plate on a flat surface and check the amount of deformation with a feeler gauge. Max flatness error: **0.2 mm**.



Overhaul of the pressure plate

Check the condition of the bearing: renew the bearing if the play is excessive.
Check the contact surfaces of the last friction plate; If extremely scored, polish it in the same manner as described previously for the cylinder head surface.
Check the condition of the spring guide cups and always replace the spring retaining collars.

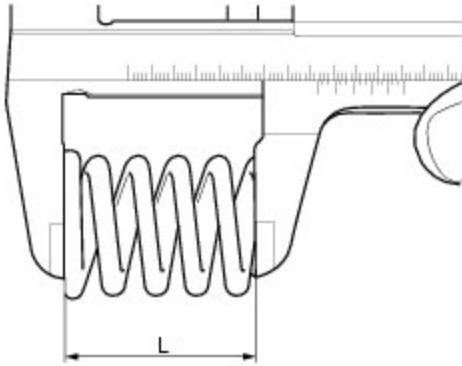


Overhauling the pressure plate springs

Measure the length "L" of each spring (3):

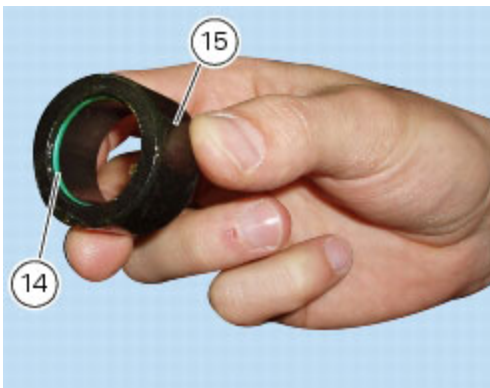
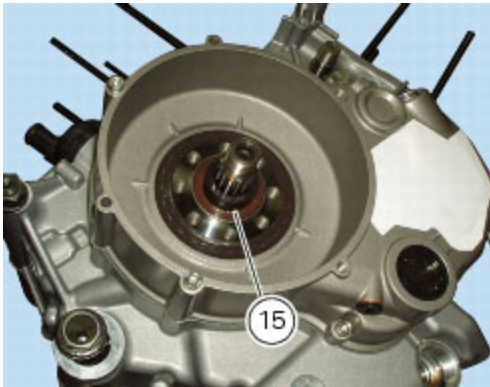
Minimum length: **41** mm.

Renew any springs that are shorter than the above limit value.

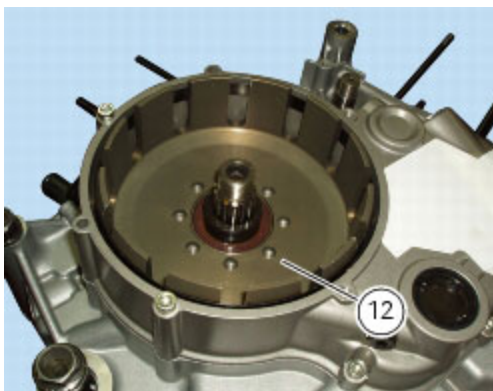


Reassembling the clutch

Fit the spacer (15) to the gearbox primary shaft along with the lubricated O-ring (14). The side with the O-ring should face the engine.



Assemble the clutch drum (12) to the primary drive gear.



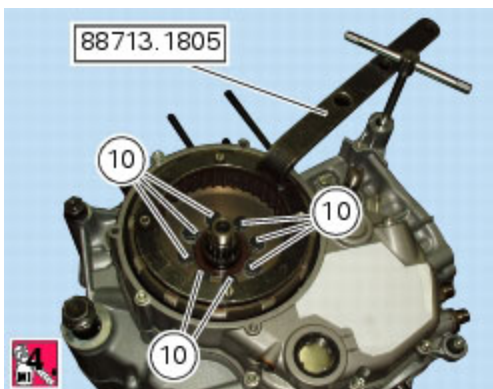
Apply threadlocker to the threads of the eight retaining screws (10) and insert them into the gear.



Important

Use new retaining screws for the clutch drum.

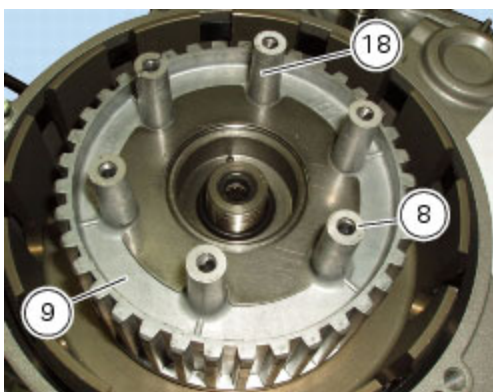
Use tool no. **88713.1805** to lock out the clutch drum rotation and tighten the screws (10) to the specified torque, in a crosswise sequence (Sect. C 3, [Engine torque settings](#)).

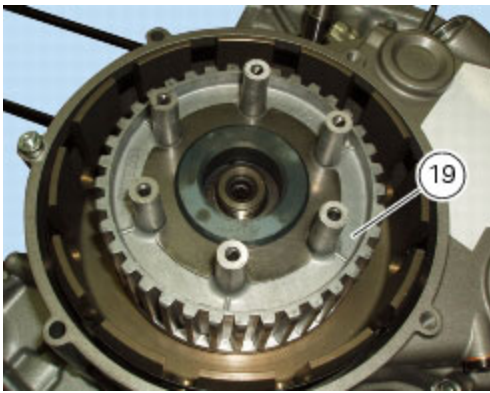


Fit the clutch centre (9) complete with the cush drive bushes onto the primary shaft.

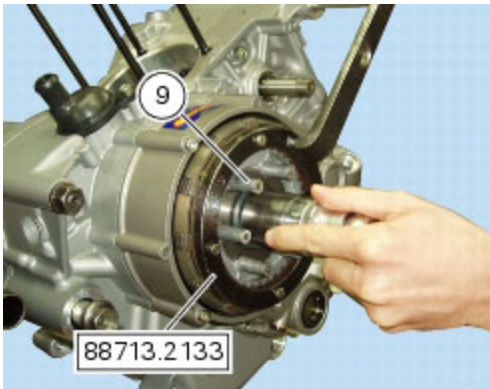
Fit the following parts onto the end of the primary shaft:

- thrust washer (8);
- bush (19) with pin (18) in the cush drive hub;
- nut (17).



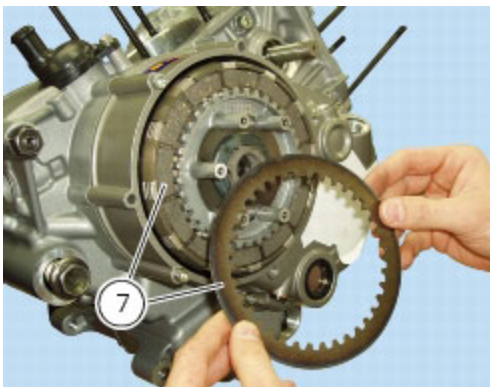
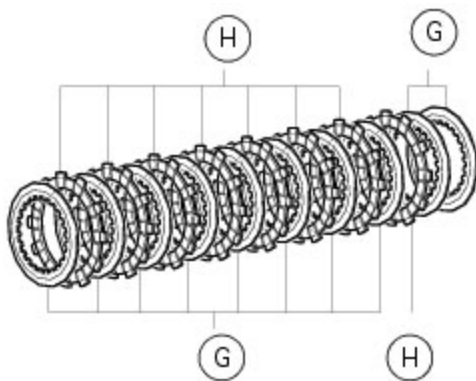


Lock the clutch drum (9) using the appropriate tool **88713.2133** and tighten the fixing nut to the specified torque (Sect. C 3, [Engine torque settings](#)).

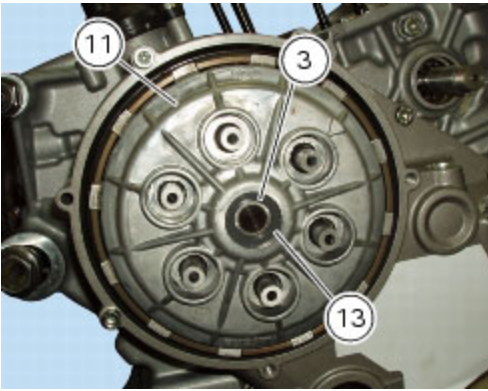


Fit the clutch plates (7) in the following order (see figure):

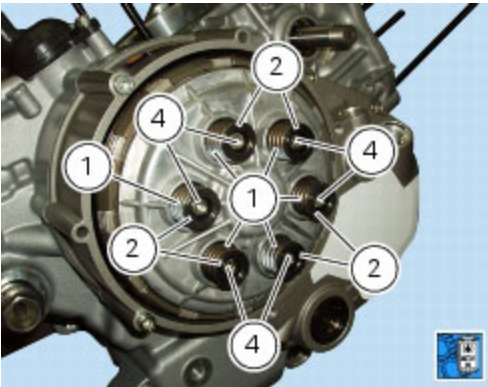
- two plain plates (G) **1.5** mm thick;
- one friction plate (H) **3** mm thick;
- a set of eight plain plates (G) alternated with eight friction plates (H). The clutch pack should be completed with a plain plate (G).



Insert the clutch lifter (3) in the bearing (13) of the pressure plate (11).
Install the pressure plate (11) so that the mark on the end of one of the clutch spring posts is aligned with the marks on the corresponding pressure plate hole, indicated by the triangle.



Fit a spring (1) and collar (2) into each recess and fit the retaining screws (4) after lubricating their threads with engine oil.
Tighten the screws (4) progressively in a crosswise pattern and then tighten to the specified torque (Sect. C 3, [Engine torque settings](#)).



Locate the gasket (20) and the clutch cover (21) and then tighten the screws (23) and (22) to the specified torque (Sect. C 3, [Engine torque settings](#)).

